```
> restart;
    #The Wave Equation Scale (where hbar noted as h because is more
    covenable);20 Feb.2026
```

```
> eq6 := 2*m/h^2*(E + ec^2/rho - L^2/(2*m*rho^2) - h^2/(2*m)*k)*rho^2/R*f(rho) + diff(rho*f(rho), rho, rho);#Onduscular;
```

$$eq6 := \frac{2\,m\left(E+\frac{ec^2}{\rho}-\frac{L^2}{2\,m\,\rho^2}-\frac{h^2\,k}{2\,m}\right)\rho^2 f(\rho)}{h^2\,R}+2\,\frac{\mathrm{d}}{\mathrm{d}\rho}f(\rho)+\rho\left(\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}f(\rho)\right) \tag{1}$$

```
> change_of_vars := { rho = rb*r, f(rho) = g(r)};
```

$$change_of_vars := \{\rho = rb\,r, f(\rho) = g(r)\} \tag{2}$$

```
> PDEtools:-dchange(change_of_vars, eq6, [g(r), r]);
```

$$\frac{2\,m\left(E+\frac{ec^2}{rb\,r}-\frac{L^2}{2\,m\,rb^2\,r^2}-\frac{h^2\,k}{2\,m}\right)rb^2\,r^2\,g(r)}{h^2\,R}+\frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)}{rb}+\frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)}{rb} \tag{3}$$

```
> eq1 := 2*m*(E + ec^2/(rb*r) - L^2/(2*m*rb^2*r^2) - h^2/(2*m)*k)*rb^2*r^2*g(r)/(h^2*R) + 2*(diff(g(r), r))/rb
    + r*(diff(g(r), r, r))/rb;
```

$$eq1 := \frac{2\,m\left(E+\frac{ec^2}{rb\,r}-\frac{L^2}{2\,m\,rb^2\,r^2}-\frac{h^2\,k}{2\,m}\right)rb^2\,r^2\,g(r)}{h^2\,R}+\frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)}{rb}+\frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)}{rb} \tag{4}$$

```
> subs(E = m*ec^4/(32*Pi^2*epsilon^2*h^2)*E0, rb = 4*Pi*epsilon*h^2/(m*ec^2), eq1);
```

$$\frac{32\,h^2\left(\frac{m\,ec^4\,E0}{32\,\pi^2\,\epsilon^2\,h^2}+\frac{ec^4\,m}{4\,\pi\,\epsilon\,h^2\,r}-\frac{L^2\,m\,ec^4}{32\,\pi^2\,\epsilon^2\,h^4\,r^2}-\frac{h^2\,k}{2\,m}\right)\pi^2\,\epsilon^2\,r^2\,g(r)}{m\,ec^4\,R}+\frac{\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)m\,ec^2}{2\,\pi\,\epsilon\,h^2}+\frac{m\,ec^2\,r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)}{4\,\pi\,\epsilon\,h^2} \tag{5}$$

```
> simplify(%);
```

$$\frac{1}{4\,h^2\,\pi\,\epsilon\,m^2\,ec^4\,R}\left(m^3\,ec^6\,r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)R+2\left(\frac{\mathrm{d}}{\mathrm{d}r}g(r)\right)m^3\,ec^6\,R-4\,\pi\,\epsilon\left(m^2\left(-E0\,h^2\,r^2-8\,\pi\,\epsilon\,h^2\,r+L^2\right)ec^4+16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)g(r)\right) \tag{6}$$

```
> r*(diff(g(r), r, r))+2*diff(g(r), r)-4*g(r)*(m^2*(-E0*h^2*r^2 - 8*Pi*
  epsilon*h^2*r + L^2)*ec^4 + 16*epsilon^2*h^6*k*r^2*Pi^2)*epsilon*Pi*m*
  ec^2/(4*h^2*Pi);
```

$$r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}g(r)\right)+2\,\frac{\mathrm{d}}{\mathrm{d}r}g(r)-\frac{g(r)\left(m^2\left(-E0\,h^2\,r^2-8\,\pi\,\epsilon\,h^2\,r+L^2\right)ec^4+16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)\epsilon\,m\,ec^2}{h^2} \tag{7}$$

```
> simplify((-m^2*(-E0*h^2*r^2 - 8*Pi*epsilon*h^2*r + L^2)*ec^4 + 16*
  epsilon^2*h^6*k*r^2*Pi^2)*epsilon*m*ec^2/h^2);
```

$$-\frac{ec^2\left(\left(\left(-E0\,r^2-8\,\pi\,\epsilon\,r\right)h^2+L^2\right)m^2\,ec^4-16\,\epsilon^2\,h^6\,k\,r^2\,\pi^2\right)\epsilon\,m}{h^2} \tag{8}$$

```
> eq2 := (E0 + 8·Pi·ϵ/r - L^2/(h^2·r^2) - 16·h^4·k·π^2·ϵ^2/(m^2·ec^4))·r^2/R·g(r) + diff(r·g(r), r, r);
```

$$eq2 := \frac{\left(E0+\frac{8\,\pi\,\epsilon}{r}-\frac{L^2}{h^2\,r^2}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4}\right)r^2\,g(r)}{R}+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{9}$$

```
>
```

```
> eq2 := (E0 + 8 π ϵ/r - L^2/(h^2·r^2) - 16 h^4 k π^2 ϵ^2/(m^2 ec^4))·r^2/R ·g(r) + 2 d/dr g(r) + r (d^2/dr^2 g(r));
```

$$eq2 := \frac{\left(E0+\frac{8\,\pi\,\epsilon}{r}-\frac{L^2}{h^2\,r^2}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4}\right)r^2\,g(r)}{R}+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{10}$$

```
> TFreeOrd := (E0/R + 8 π ϵ/(r·R) - L^2/(h^2 r^2·R) - 16 h^4 k π^2 ϵ^2/(m^2 ec^4·R))·r^2·g(r);
```

$$TFreeOrd := \left(\frac{E0}{R}+\frac{8\,\pi\,\epsilon}{r\,R}-\frac{L^2}{h^2\,r^2\,R}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4\,R}\right)r^2\,g(r) \tag{11}$$

```
> sort(((E0/R + 8 π ϵ/(r R) - L^2/(h^2 r^2 R) - 16 h^4 k π^2 ϵ^2/(m^2 ec^4 R)) r^2), r);
```

$$\left(\frac{8\,\pi\,\epsilon}{R\,r}-\frac{L^2}{h^2\,R\,r^2}+\frac{E0}{R}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2}{m^2\,ec^4\,R}\right)r^2 \tag{12}$$

```
> eq2 := (E0 r^2/R - 16 h^4 k π^2 ϵ^2 r^2/(m^2 ec^4 R) + 8 π ϵ r/R - L^2/(h^2 R))·g(r) + 2 d/dr g(r) + r (d^2/dr^2
    g(r));#Onduscular equation.
```

$$eq2 := \left(\frac{E0\,r^2}{R}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2\,r^2}{m^2\,ec^4\,R}+\frac{8\,\pi\,\epsilon\,r}{R}-\frac{L^2}{h^2\,R}\right)g(r)+2\,\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)+r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right) \tag{13}$$

```
> restart;
    #ScrodingeScal where hbar noted as h because is more covenable;we
    start from eqation 19.8 Feyman
```

```
> eq1 := 2 m/h^2·(E + ec^2/ρ)·ρ·f(ρ) + diff(ρ·f(ρ), ρ, ρ);#Schrodinger;
```

$$eq1 := \frac{2\,m\left(E+\frac{ec^2}{\rho}\right)\rho\,f(\rho)}{h^2}+2\,\frac{\mathrm{d}}{\mathrm{d}\rho}\,f(\rho)+\rho\left(\frac{\mathrm{d}^2}{\mathrm{d}\rho^2}\,f(\rho)\right) \tag{14}$$

```
> change_of_vars := { rho = rb·r, f(rho) = g(r)};
```

$$change_of_vars := \{\rho = rb\,r, f(\rho) = g(r)\} \tag{15}$$

```
> PDEtools:-dchange(change_of_vars, eq1, [g(r), r]);
```

$$\frac{2\,m\left(E+\frac{ec^2}{rb\,r}\right)rb\,r\,g(r)}{h^2}+\frac{2\left(\frac{\mathrm{d}}{\mathrm{d}r}\,g(r)\right)}{rb}+\frac{r\left(\frac{\mathrm{d}^2}{\mathrm{d}r^2}\,g(r)\right)}{rb} \tag{16}$$

```
> eq2 := 2 m (E + ec^2/(rb r)) rb^2 r g(r)/h^2 + d^2/dr^2( r·g(r));#we multiply rb ;
```

$$eq2 := \frac{2\,m\left(E+\frac{ec^2}{rb\,r}\right) rb^2\,r\,g(r)}{h^2} + 2\,\frac{d}{dr}\,g(r) + r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{17}$$

```
> subs(E = m·ec^4/(32·Pi^2·ϵ^2·h^2)·E0, rb = 4·Pi·ϵ· h^2/(m·ec^2), eq2);
    #with wacuum permittivity ϵ and E0=energy in Rythberg unit
```

$$\frac{32\left(\frac{m\,ec^4\,E0}{32\,\pi^2\,\epsilon^2\,h^2}+\frac{ec^4\,m}{4\,\pi\,\epsilon\,h^2\,r}\right)\pi^2\,\epsilon^2\,h^2\,r\,g(r)}{m\,ec^4} + 2\,\frac{d}{dr}\,g(r) + r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{18}$$

```
> eq2 := 32 ·(E0/(32· h^2) + Pi·ϵ/(4· h^2 ·r)) ·h^2 ·r ·g(r) + 2 d/dr g(r) + r (d^2/dr^2 g(r));
```

$$eq2 := 32\left(\frac{E0}{32\,h^2}+\frac{\pi\,\epsilon}{4\,h^2\,r}\right)h^2\,r\,g(r) + 2\,\frac{d}{dr}\,g(r) + r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{19}$$

```
> eq2 := 32 (E0/32 + π ϵ/(4· r))· r ·g(r) + 2 d/dr g(r) + r (d^2/dr^2 g(r));
```

$$eq2 := 32\left(\frac{E0}{32}+\frac{\pi\,\epsilon}{4\,r}\right)r\,g(r) + 2\,\frac{d}{dr}\,g(r) + r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{20}$$

```
> eq2 := (E0 + 8·π·ϵ/r)· r ·g(r) + d^2/dr^2( r·g(r));
```

$$eq2 := \left(E0+\frac{8\,\pi\,\epsilon}{r}\right)r\,g(r) + 2\,\frac{d}{dr}\,g(r) + r\left(\frac{d^2}{dr^2}\,g(r)\right) \tag{21}$$

```
> restart;
    # we have the Schrodinger Scaled equation with wacuum permittivity
    ϵ
> eq3 := (E0 + 8·π·ϵ/ρ)·ρ·f(ρ) + d^2/dρ^2(ρ·f(ρ));
    #eqation (1) with wacuum permittivity ϵ
```

$$eq3 := \left(E0+\frac{8\,\pi\,\epsilon}{\rho}\right)\rho\,f(\rho) + 2\,\frac{d}{d\rho}\,f(\rho) + \rho\left(\frac{d^2}{d\rho^2}\,f(\rho)\right) \tag{22}$$

```
> restart;
    #now we have the Schrodinger Scaled eq1 with wacuum permittivity ϵ;
> eq1 := (E0 + 8·π·ϵ/ρ)· ρ ·f(ρ) + d^2/dρ^2(ρ·f(ρ));
```

$$eq1 := \left(E0+\frac{8\,\pi\,\epsilon}{\rho}\right)\rho\,f(\rho) + 2\,\frac{d}{d\rho}\,f(\rho) + \rho\left(\frac{d^2}{d\rho^2}\,f(\rho)\right) \tag{23}$$

```
> change_of_vars := { rho = rb·r, f(rho) = g(r)·(R − r·rb)/R};
```

$$change_of_vars := \left\{\rho = rb\,r,\ f(\rho) = \frac{g(r)\ (-rb\,r+R)}{R}\right\} \tag{24}$$

```
> PDEtools:-dchange(change_of_vars, eq1, [g(r), r]);
```

$$\frac{\left(E0+\frac{8\,\pi\,\epsilon}{rb\,r}\right)rb\,r\,g(r)\,(-rb\,r+R)}{R}+\frac{2\left(\frac{\left(\frac{d}{dr}\,g(r)\right)(-rb\,r+R)}{R}-\frac{g(r)\,rb}{R}\right)}{rb}+\frac{r\left(\frac{\left(\frac{d^2}{dr^2}\,g(r)\right)(-rb\,r+R)}{R}-\frac{2\left(\frac{d}{dr}\,g(r)\right)rb}{R}\right)}{rb} \tag{25}$$

```
> eq2 := simplify(%);
```

$$eq2 := \frac{1}{rb\,R}\left(r\,(-rb\,r+R)\left(\frac{d^2}{dr^2}\,g(r)\right)+(-4\,rb\,r+2\,R)\left(\frac{d}{dr}\,g(r)\right)+g(r)\,(-r^2\,E0\,rb^2+r\,(E0\,R-8\,\pi\,\epsilon)\,rb+8\,R\,\pi\,\epsilon-2)\,rb\right) \tag{26}$$

```
> eq2 := %/(R-r*rb);#we devide with (-r rb+ R );
```

$$eq2 := \frac{1}{rb\,R\,(-rb\,r+R)}\left(r\,(-rb\,r+R)\left(\frac{d^2}{dr^2}\,g(r)\right)+(-4\,rb\,r+2\,R)\left(\frac{d}{dr}\,g(r)\right)+g(r)\,(-r^2\,E0\,rb^2+r\,(E0\,R-8\,\pi\,\epsilon)\,rb+8\,R\,\pi\,\epsilon-2)\,rb\right) \tag{27}$$

```
>
> eq2 := r/(rb*R)*(diff(g(r),r$2))+(-4*rb*r+2*R)/(rb*R*(-r*rb+R))*(diff(g(r),r))+g(r)/(R*(-r*rb+R))*(
    -r^2*E0*rb^2+r*(E0*R-8*Pi*epsilon)*rb+8*R*Pi*epsilon-2);
    #We aproximate (-4*rb*r+4*R)/(-r*rb+2*R)~2;
```

$$eq2 := \frac{r\left(\frac{d^2}{dr^2}\,g(r)\right)}{rb\,R}+\frac{(-4\,rb\,r+2\,R)\left(\frac{d}{dr}\,g(r)\right)}{rb\,R\,(-rb\,r+R)}+\frac{g(r)\,(-r^2\,E0\,rb^2+r\,(E0\,R-8\,\pi\,\epsilon)\,rb+8\,R\,\pi\,\epsilon-2)}{R\,(-rb\,r+R)} \tag{28}$$

```
> eq2 := r*(diff(g(r),r$2))+2*diff(g(r),r)
    +g(r)*rb*(-r^2*E0*rb^2+r*(E0*R+8*Pi*epsilon)*rb+8*R*Pi*epsilon-2)/(-rb*r+R);
    #We multiply with rb*R
```

$$eq2 := r\left(\frac{d^2}{dr^2}\,g(r)\right)+2\,\frac{d}{dr}\,g(r)+\frac{g(r)\,rb\,(-r^2\,E0\,rb^2+r\,(E0\,R+8\,\pi\,\epsilon)\,rb+8\,R\,\pi\,\epsilon-2)}{-rb\,r+R} \tag{29}$$

```
> TFreeOrd := simplify(rb*(-r^2*E0*rb^2+r*(E0*R+8*Pi*epsilon)*rb+R*Pi*epsilon-2)/(-rb*r+R));
```

$$TFreeOrd := \frac{rb\,(-r^2\,E0\,rb^2+r\,(E0\,R+8\,\pi\,\epsilon)\,rb+R\,\pi\,\epsilon-2)}{-rb\,r+R} \tag{30}$$

```
> series(rb*(-r^2*E0*rb^2+r*(E0*R+8*Pi*epsilon)*rb+8*R*Pi*epsilon-2)/(-rb*r+R), r);
```

$$\frac{rb\,(8\,R\,\pi\,\epsilon-2)}{R}+\frac{(E0\,R+8\,\pi\,\epsilon)\,rb^2+\frac{2\,rb^2\,(4\,R\,\pi\,\epsilon-1)}{R}}{R}\,r \tag{31}$$
$$+\frac{-rb^3\,E0+\frac{rb^3\,(E0\,R^2+16\,R\,\pi\,\epsilon-2)}{R^2}}{R}\,r^2+2\,\frac{rb^4\,(8\,R\,\pi\,\epsilon-1)}{R^4}\,r^3+2\,\frac{rb^5\,(8\,R\,\pi\,\epsilon-1)}{R^5}\,r^4$$
$$+2\,\frac{rb^6\,(8\,R\,\pi\,\epsilon-1)}{R^6}\,r^5+O(r^6)$$

```
> eq3 := r*(diff(g(r),r$2)) + 2*diff(g(r),r) + (rb*(+8*R*π*ϵ-2)/R + rb
      *((E0*R+8*π*ϵ)*rb^2 - 2*(4*R*π*ϵ+1)*rb^2/R)/R*r + (-rb^3*E0 + rb^3*(E0*R^2-2)/R^2)/R*r^2)
      *g(r);#Aproximative
```

$$eq3 := r\left(\frac{d^2}{dr^2}\,g(r)\right)+2\,\frac{d}{dr}\,g(r)+\left(\frac{rb\,(8\,R\,\pi\,\epsilon-2)}{R}\right. \tag{32}$$
$$\left.+\frac{rb\left((E0\,R+8\,\pi\,\epsilon)\,rb^2-\frac{2\,(4\,R\,\pi\,\epsilon+1)\,rb^2}{R}\right)r}{R}+\frac{\left(-rb^3\,E0+\frac{rb^3\,(E0\,R^2-2)}{R^2}\right)r^2}{R}\right)$$
$$g(r)$$

```
> rb*(+8*R*π*ϵ-2)/R + rb*((E0*R+8*π*ϵ)*rb^2 - 2*(4*R*π*ϵ+1)*rb^2/R)/R*r
      + ((-rb^3*E0 + rb^3*(E0*R^2-2)/R^2)/R)*r^2 = E0*r^2/R - 16*h^4*k*π^2*ϵ^2*r^2/(m^2*ec^4*R) - 8*π*ϵ*r/R
      + L^2/(h^2*R);#Aproximation;
```

$$\frac{rb\,(8\,R\,\pi\,\epsilon-2)}{R}+\frac{rb\left((E0\,R+8\,\pi\,\epsilon)\,rb^2-\frac{2\,(4\,R\,\pi\,\epsilon+1)\,rb^2}{R}\right)r}{R} \tag{33}$$
$$+\frac{\left(-rb^3\,E0+\frac{rb^3\,(E0\,R^2-2)}{R^2}\right)r^2}{R}=\frac{E0\,r^2}{R}-\frac{16\,h^4\,k\,\pi^2\,\epsilon^2\,r^2}{m^2\,ec^4\,R}-\frac{8\,\pi\,\epsilon\,r}{R}+\frac{L^2}{h^2\,R}$$

```
> solve({(-rb^3*E0 + rb^3*(E0*R^2-2)/R^2)/R = E0/R - 16*h^4*k*π^2*ϵ^2/(m^2*ec^4*R),
      (rb*(2*E0*R*rb - 8*π*ϵ*rb) + rb^2*(8*R*π*ϵ-1)/R)/(2*R) = -8*π*ϵ/R,
```

```
        rb (8 R π ϵ − 2)          L^2
       ----------------- = - ------------ }, {R, k, E0} );
              R                h^2 R
```

$$\left\{E0 = \frac{32\,\pi^2\,\epsilon^2\,h^2\left(h^2\,rb^3 - 4\,rb\,h^2 + 2\,L^2\right)}{rb\left(4\,h^4\,rb^2 - 4\,L^2\,h^2\,rb + L^4\right)},\ R = -\frac{-2\,rb\,h^2 + L^2}{8\,\pi\,\epsilon\,h^2\,rb},\ k\right. \tag{34}$$

$$\left. = \frac{2\,m^2\,ec^4\left(4\,h^2\,rb^6 + h^2\,rb^3 - 4\,rb\,h^2 + 2\,L^2\right)}{rb\,h^2\left(4\,h^4\,rb^2 - 4\,L^2\,h^2\,rb + L^4\right)}\right\}$$

```
> R := (2 rb h^2-L^2)/(8 π ϵ h^2 rb);
```

$$R := \frac{2\,rb\,h^2 - L^2}{8\,\pi\,\epsilon\,h^2\,rb} \tag{35}$$

```
> subs( rb = (4·Pi·ϵ· h^2)/(m·ec^2), % );
```

$$\frac{\left(\frac{8\,\pi\,\epsilon\,h^4}{m\,ec^2} - L^2\right) m\,ec^2}{32\,\pi^2\,\epsilon^2\,h^4} \tag{36}$$

```
> simplify(%);#The R value
```

$$\frac{8\,\pi\,\epsilon\,h^4 - L^2\,m\,ec^2}{32\,\pi^2\,\epsilon^2\,h^4} \tag{37}$$

```
> #We substitute according convetion h with hbar
> m := 9.10938356*10^(-31); epsilon := 8.8541878128*10^(-12);hbar :=
  6.62607015/(2*Pi)*10^(-34);
  ec := 1.602176*10^(-19); c := 299792458; h := 6.62607015*10^(-34); rb
  := 5.292410932*10^(-11);L:=(21*22)^0.5*hbar;metru := 1.057023410*10^
  (-16);l:=21;
```

$$m := 9.109383560 \times 10^{-31}$$
$$\epsilon := 8.854187813 \times 10^{-12}$$
$$\hbar := 1.054571818 \times 10^{-34}$$
$$ec := 1.602176000 \times 10^{-19}$$
$$c := 299792458$$
$$h := 6.626070150 \times 10^{-34}$$
$$rb := 5.292410932 \times 10^{-11}$$
$$L := 2.266716203 \times 10^{-33}$$
$$metru := 1.057023410 \times 10^{-16}$$
$$l := 21 \tag{38}$$

```
> R:=-(8*Pi*epsilon*hbar^4 - L^2*ec^2*m)/(32*Pi^2*epsilon^2*hbar^4);R*
  metru;R := (-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^2*rb);# The R value
  in m and in LightYears;
```

$$R := 3.923303466 \times 10^{22}$$
$$4.147023608 \times 10^{6}$$
$$R := 3.922833075 \times 10^{22} \tag{39}$$

```
> R:=l*(l+1)/(8*Pi*epsilon*rb);R*metru;
```

$$R := 3.922833072 \times 10^{22}$$

$$4.146526391 \times 10^{6} \tag{40}$$

```
> L:=(21*22)^0.5*hbar;(-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^2*rb);%*
  metru;
```

$$L := 2.266716203 \times 10^{-33}$$

$$3.922833075 \times 10^{22}$$

$$4.146526394 \times 10^{6} \tag{41}$$

```
> (-hbar^2*rb + L^2)/(8*Pi*epsilon*hbar^2*rb)
```

$$3.922833075 \times 10^{22} \tag{42}$$

```
> l*(l+1)/(8*rb*Pi*epsilon);
```

$$3.922833072 \times 10^{22} \tag{43}$$